

$$3^x = 3^{5+x}$$

$$x = 5+x$$

$$\frac{-6x}{-6} = \frac{5}{-6}$$

$$2^x + 9 \cdot 2^x = 40$$

$$x+x=40$$

assolutamente non posso farlo

Raccogliendo 2^x (esponenziale)

$$2^x(-1+9) = 40$$

$$\frac{2^x \cdot -10 = 40}{-10} = \frac{40}{-10} \quad 4$$

$$2^x = 4$$

$$2^x = 2^2$$

$$x = 2$$

③ TIPO DI RACCOLGIMENTO

$$45 \cdot 25^{x-1} - 5^{2x+1} = 50$$

$$45 \cdot 25^x \cdot 25^{-1} - 5^{2x} \cdot 5^1 = 50$$

$$45 \cdot 5^{2x} \cdot \frac{1}{5} - 5^{2x} \cdot 5^1 = 50$$

$$3 \cdot 5^{2x} - 5^{2x} \cdot 5^1 = 50$$

$$5^{2x}(3-5) = 50$$

$$\frac{5^{2x} \cdot (-2) = 50}{5^{2x} = 5^3} = \frac{-50}{-2} \quad 25$$

$$\frac{2x}{2} = \frac{3}{2}$$

$$x = \frac{3}{2}$$

③ UTILIZZO DI UN'INCOGNITA AUSILIARIA

$$3^{2x} - 3 \cdot 3^x - 4 = 0$$

$$t^2 - 3t - 4 = 0$$

$t = 3^x$
 $t^2 = (3^x)^2 = 3^{2x}$

$$\Delta = 9 - 4(-4)$$

$$= 9 + 16 = 25$$

$$x_{1,2} = \frac{3 \pm 5}{2}$$

$$t = 4 \quad t = -1$$

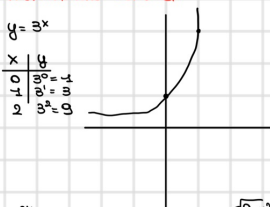
$$3^x = 4$$

non posso andare avanti

$$3x = -1$$

impossibile

PREPARAZIONE VERIFICA



$$5^{2x} = \frac{1}{25}$$

$$5^{2x} = \frac{1}{5^2}$$

$$5^{2x} = 5^{-2}$$

$$\frac{2x}{2} = \frac{-2}{2}$$

$$x = -1$$

$$x = -\frac{1}{3}$$

$y = 4\sqrt{9-x^2}$

$$9 - x^2 > 0$$

$$-x^2 + 9 > 0$$

$$\frac{-x^2}{-1} > \frac{-9}{-1}$$

$$x^2 < 9$$

$$x^2 = \pm 3$$

$-3 \leq x \leq 3$

$$2x \geq 1$$

$$\frac{4^2 - x^2 \cdot 2^{x+3}}{16^x} = \frac{1}{8}$$

$$\frac{4^{4-2x} \cdot 2^{x+3}}{2^{4x}} = 2^{-3}$$

$$\frac{2^{4-2x+x+3}}{2^{4x}} = 2^{-3}$$

$$\frac{2^{-x+7}}{2^{4x}} = 2^{-3}$$

$$2^{5x+7} = 2^{-3}$$

$$-5x+7 = -3$$

$$\frac{-5x}{-5} = \frac{-10}{-5}$$

$$x = 2$$

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$$4^{2x+1} = 8^{2x-1}$$

$$4^{2x+1} = 2^{6x-3}$$

$$2^{4x+2} = 2^{6x-3}$$

$$4x+2 = 6x-3$$

$$\frac{-2x}{-2} = \frac{-5}{-2}$$

$$x = 5/2$$

$y = 4\left(\frac{1}{4}\right)^x$

x	y
0	$\left(\frac{1}{4}\right)^0 = 1$
1	$\left(\frac{1}{4}\right)^1 = \frac{1}{4}$
2	$\left(\frac{1}{4}\right)^2 = \frac{1}{16}$

$$\frac{3 \cdot 5^x + 5^{x+1}}{3 \cdot 5^x + 5^x \cdot 5^1} = \frac{8 \cdot 5^3}{8 \cdot 5^3}$$

$$\frac{5^x(3+5)}{5^x(3+5)} = \frac{8 \cdot 5^3}{8 \cdot 5^3}$$

$$\frac{5^x \cdot 8}{8 \cdot 5^3} = \frac{8 \cdot 5^3}{8 \cdot 5^3}$$

$$5^x = 5^3$$

$$x = 3$$

$$\left(\frac{1}{4}\right)^x = -2$$

impossibile perché non posso mai avere un numero negativo.

$$y = 5^{\frac{x+3}{x-2}}$$

$$x^2 - x \neq 0$$

$$x(x-1) \neq 0$$

$$x \neq 0 \quad x \neq 1$$